

CHER SCARLETT

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EDUCATION

2026 **ORANGE COAST COLLEGE** GPA: 4.0

May | A.S., Physics, Mathematics, Astronomy, & Natural Science; A.A. Liberal Arts

2024 **Extracurriculars:** Honors, Astronomy Club (*Treasurer*), STEM Club (*Outreach Manager*),
January MESA, EOPS, Math & Science Club (*President*), $\Phi\Theta\K$, NLSL, SXS group
Professional Memberships: Institute of Physics, Society of Physics Students, American Physical Society, American Geophysical Union, American Mathematical Society, American Association of Physics Teachers, American Meteorological Society

Includes enrollment in other California community colleges.

SKILLS

Python | Java | Groovy | Ruby | PHP | C | JavaScript | CSS | HTML | IDL | Fortran
LaTeX | Mathematica | MATLAB | Jupyter Notebooks | AstroImageJ | ArcGIS | PyTorch

RESEARCH EXPERIENCES

2026 **CUESTA COLLEGE**

Double Star Astrometry | Advisor: Dr. James Eickemeyer

2025 **INSTITUTE FOR STUDENT ASTRONOMICAL RESEARCH (INSTAR)**

Las Cumbres Observatory (LCO) Network

Astronomy Research Seminar | Advisor: Dr. Rachel Freed

2025 **NASA JET PROPULSION LABORATORY (NASA-JPL)**

NASA Juno Mission

Analysis and Archiving of Near- and Mid-Infrared Observations of Outer Planet Atmospheres

Planetary and Exoplanetary Atmospheres Group

Student Independent Research Internship (SIRI) | Advisor: Dr. Glenn S. Orton

2025 **CALIFORNIA INSTITUTE OF TECHNOLOGY (CALTECH)**

| *NASA-JPL Lunar Trailblazer Mission*

2024 *Planetary Sciences Department*

Research Assistant, Science Communication Intern | Advisor: Dr. Bethany Ehlmann

2025 **ORANGE COAST COLLEGE**

Compact Objects and Gravitational Waves | Independent Study | Advisor: Prof. Jessica Asbell

ACADEMIC & WORKSHOP EXPERIENCES

2025 **NASA AI/ML SCIENCE AND TECHNOLOGY INTEREST GROUP**

PyTorch and Auto Differentiation Lecture

2025 **COMPUTATIONAL RESEARCH ACCESS NETWORK | BRYN MAWR | NSF**

Computational Research Methods for Plasma Physics & Sciences Virtual Seminar

Magnetohydrodynamics with FLASH, Monte Carlo Method, & Particle-in-Cell Tracks

2024 **UNIVERSITY OF ILLINOIS | GRAVITY THEORY TRUST**

EFTs, Gravity, and Fluid Dynamics: Progress, Challenges, and Emerging Opportunities Workshop

- 2024 CALTECH | NASA EXOPLANET SCIENCE INSTITUTE (NEXSCI)**
Sagan Exoplanet Summer Workshop
Advances in Direct Imaging: From Young Jupiters to Habitable Earths
- 2024 MIAMI UNIVERSITY | NSF | EARTHSCOPE CONSORTIUM**
Seismology Skill Building Workshop
- 2024 QUEEN'S UNIVERSITY | CERN**
The 6th Summer Particle Astrophysics Workshop (EIEIOO)
- 2024 ARIZONA STATE UNIVERSITY | NASA**
Lucy Student Pipeline Accelerator and Competency Enabler (L'SPACE) Academy
NASA Proposal Writing and Evaluation Experience (NPWEE)
- 2024 NASA**
NASA Community College Aerospace Scholars (NCAS)
- 2003 FRED HUTCHINSON CANCER CENTER | UNIVERSITY OF WASHINGTON**
HutchLab | High School Human Genome Project

OBSERVATION EXPERIENCES

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|-------------|---|-----------------------------|
| 2025 | Caltech Palomar Observatory - Hale Telescope
<i>4 nights; 5.1 m; 1 - 2.5 μm (PHARO)</i>
<i>Observing Jupiter in Support of Juno Mission (NASA-JPL)</i> | <i>Palomar Mountain, CA</i> |
| 2025 | Las Cumbres Observatory (LCO) - Teide Observatory
<i>2 nights; 0.35 m; 0.4 - 0.7 μm (QHY600)</i>
<i>Astrometry of double stars</i> | <i>Tenerife, ES</i> |
| 2025 | LCO - South African Astronomical Observatory
<i>1 nights; 0.35 m; 0.4 - 0.7 μm (QHY600)</i>
<i>Astrometry of double stars</i> | <i>Sutherland, ZA</i> |
| 2025 | NASA Infrared Telescope Facility (IRTF)
<i>10 nights; 3.2 m; 0.8 - 5.3 μm (iSHELL, SPEX, & MIRSI)</i>
<i>Observing Jupiter in Support of Juno Mission; Volatiles in Comet Comae</i> | <i>Mauna Kea, HI</i> |
| 2025 | Lockheed Martin Ignite Ground System (IGS)
<i>4 nights; 8-9 kHz</i>
<i>Monitoring for Lunar Trailblazer (LTB) Signal</i> | <i>Waterton, CO</i> |
| 2025 | Fukui University of Technology (FUT) Awara Space Center
<i>2 nights; 8-9 kHz</i>
<i>Monitoring for Lunar Trailblazer (LTB) Signal</i> | <i>Fukui, JP</i> |
| 2025 | Australia Telescope National Facility
<i>6 nights; 8-9 kHz (Mopra)</i>
<i>Monitoring for Lunar Trailblazer (LTB) Signal</i> | <i>New South Wales, AU</i> |

TEACHING & TUTORING EXPERIENCES

2025	EARTHSCOPE CONSORTIUM MIAMI UNIVERSITY	<i>Miami, OH</i>
September	<i>Teaching Assistant</i>	
	<i>Seismology Skill Building Workshop</i>	
May		
2025	ORANGE COAST COLLEGE PLANETARIUM	<i>Costa Mesa, CA</i>
May	<i>Presenter</i>	
2024		
May		
2018	UNIVERSITY OF WASHINGTON	<i>Seattle, WA</i>
May	<i>Adjunct Instructor</i>	
	<i>Introductory Web Development (Extension)</i>	
January		

ENGINEERING EXPERIENCES

2023	CONTROLZEE, INC	<i>San Jose, CA</i>
April	<i>Senior Software Engineer</i>	
2022		
March		
2021	APPLE, INC	<i>Cupertino, CA</i>
November	<i>Principal Software Engineer, Global Security Tools</i>	
2020		
April		
2020	WEBFLOW	<i>San Francisco, CA</i>
April	<i>Senior Software Engineer, Design System & Core Engine</i>	
2019		
April		
2019	STARBUCKS COFFEE COMPANY	<i>Seattle, WA</i>
April	<i>Lead Software Engineer, Web & Mobile</i>	
2017		
November		
2017	WORLD WIDE TECHNOLOGY	<i>St. Louis, MO</i>
August	<i>Senior Software Engineer</i>	
2016		
August		
2016	ACTIVISION BLIZZARD	<i>Irvine, CA</i>
August	<i>Software Engineer, Web & Mobile</i>	
2015		
August		

2015 August 2011 November	USA TODAY <i>Senior Developer, New and Emerging Technologies</i>	<i>McLean, VA</i>
2011 November 2008 January	NEUTRINO, INC <i>Front End Developer</i>	<i>Seattle, WA</i>
2008 January 2007 May	LUXURYREALESTATE.COM <i>Front End Developer</i>	<i>Seattle, WA</i>

VOLUNTEER EXPERIENCES

2021 October present	COWORKER SOLIDARITY FUND <i>Committee Member</i>	<i>Annapolis, MD</i>
2025 April	MESA STEM DAY <i>Workshop Assistant</i>	<i>Newport Beach, CA</i>
2024 December	MESA CAMPUS COUNCIL <i>Panelist</i>	<i>Newport Beach, CA</i>
2024 November	DEMOCRATIC NATIONAL COMMITTEE <i>Phone Bank Operator</i>	<i>Washington, DC</i>
2024 October	NEWPORT-MESA COLLEGE & CAREER FAIR <i>Booth Operator for Coastline College & MESA</i>	<i>Costa Mesa, CA</i>
2024 September	EXPLORE CALTECH <i>Booth Operator for Pasadena City College Physics & Astronomy</i>	<i>Pasadena CA</i>
2024 May	BOLSA CHICA CONSERVANCY <i>Invasive Species Removal</i>	<i>Huntington Beach, CA</i>
2024 May	SAN DIEGO COAST KEEPER <i>Water Quality Testing</i>	<i>San Diego, CA</i>
2023 August	SURFRIDER FOUNDATION <i>Beach Cleanup</i>	<i>San Clemente, CA</i>
2015 August	ORANGE COUNTY PROBATION DEPARTMENT <i>At-risk youth mentor (OC GRIP)</i>	<i>Santa Ana, CA</i>
2015 August	MINDMAKERS <i>Youth mentor & web developer</i>	<i>Pasadena, CA</i>

AWARDS & SCHOLARSHIPS

2026	NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION <i>Ernest F. Hollings Scholar</i>	<i>Silver Springs, MD</i>
2026	JACK KENT COOKE FOUNDATION <i>Transfer Scholar Semifinalist</i>	<i>Lansdowne, VA</i>
2025	COASTLINE COLLEGE <i>President's List (Fall)</i>	<i>Fountain Valley, CA</i>
2025	SANTA BARBARA CITY COLLEGE <i>President's Honor Roll (Fall)</i>	<i>Santa Barbara, CA</i>
2025	WOMEN'S INDEPENDENCE SCHOLARSHIP PROGRAM <i>Doris Buffett Independence Scholar</i>	<i>Wilmington, NC</i>
2025	AMERICAN INSTITUTE OF PHYSICS <i>TEAM-UP Together Scholar</i>	<i>College Park, MD</i>
2025	COASTLINE COLLEGE <i>Honor's List (Summer)</i>	<i>Fountain Valley, CA</i>
2025	ORANGE COAST COLLEGE <i>Garrison Fellow</i>	<i>Costa Mesa, CA</i>
2025	COMMUNITY COLLEGE LEAGUE OF CALIFORNIA <i>All-California Academic Award, First Team</i>	<i>Sacramento, CA</i>
2025	COASTLINE COLLEGE FOUNDATION <i>Associated Student Government Career & Technical Merit Scholarship</i>	<i>Fountain Valley, CA</i>
2025	COASTLINE COLLEGE FOUNDATION <i>Herman Tate Endowment</i>	<i>Fountain Valley, CA</i>
2025	JEANNETTE RANKIN FOUNDATION	<i>Athens, GA</i>
2024	<i>Jeannette Rankin National Scholar</i>	
2025	NATIONAL SCIENCE FOUNDATION	<i>Alexandria, VA</i>
2024	<i>S-STEM Scholar (Coastline College)</i>	
2024	COASTLINE COLLEGE <i>Honor's List (Fall)</i>	<i>Fountain Valley, CA</i>
2024	ORANGE COAST COLLEGE FOUNDATION <i>Peter Christian Hernandez Memorial Scholarship Award</i>	<i>Costa Mesa, CA</i>
2024	ORANGE COAST COLLEGE <i>Honor's List (Spring)</i>	<i>Costa Mesa, CA</i>
2024	GOLDEN WEST COLLEGE <i>Honor's List (Spring)</i>	<i>Huntington Beach, CA</i>
2022	HACKER NOON <i>Developer Defying Patriarchy (While Being an Awesome Developer) (Runner-up)</i>	<i>Edwards, CO</i>

PUBLICATIONS

- Scarlett, C. (2026). Finding Physicists: The vital role of Community Colleges. *The SPS Observer*, 60(1). (final draft submitted; publication scheduled for May)
- Scarlett, C. (2025). Lunar Trailblazer Interactive Model [Computer software]. *California Institute of Technology*.
<https://trailblazer.caltech.edu/spacecraft/index.html>
- Scarlett, C. (2025). Water into Doom: Balancing AI with Climate Action & Environmental Costs at the Water-Energy Nexus. *The Honors Transfer Council of California Building Bridges Research Conference*.
- Scarlett, C. (2025). The Climate and Tornadoes are Changing. Can Understanding Climate-Driven Chaos Save Lives?. *Academia*.
https://www.academia.edu/150281326/The_Climate_and_Tornadoes_are_Changing_Can_Understanding_Climate_Driven_Chaos_Save_Lives
- Scarlett, C. (2025). Bullshit at the Agora: On the Dangers of a 'Marketplace of Ideas'. *Academia*.
https://www.academia.edu/146050292/Bullshit_at_the_Agora_On_the_Dangers_of_a_Marketplace_of_Ideas
- Scarlett, C. (2025). No Sleep 'Til Payday: Class, Work, and the Price of Survival. *Academia*.
https://www.academia.edu/145993371/No_Sleep_Til_Payday_Class_Work_and_the_Price_of_Survival
- Scarlett, C. (2025). Emmy Noether: A Woman on the Edge of [Space]Time. *Academia*.
https://www.academia.edu/144259985/Emmy_Noether_A_Woman_on_the_Edge_of_Space_Time
- Scarlett, C. (2024). Man Made: Technology, Pygmalion displacement, and the patriarchal idealization of women in science fiction. *Academia*.
https://www.academia.edu/145993413/Man_Made_Technology_Pygmalion_displacement_and_the_patriarchal_idealization_of_women_in_science_fiction
- Scarlett, C. (2024). Remote seismology correlation and Mount St. Helens volcanology. *Seismology Skill Building Workshop*.
https://nbviewer.org/urls/www.iris.edu/hq/files/short_courses/2020/ssb/mt-rainier-seismicity.ipynb
- Scarlett, C. (2024, July 29). *A case of the blues: Is Overexposure to Artificial Light Causing a Mental Health Crisis?*. Esri. <https://storymaps.arcgis.com/stories/e236a32458bb4b7db87171f6a0305d8a>
- Carlson, G., Roginsky, J., & Scarlett, C. (2022, February 6). Washington can be a leader on protecting sexual harassment, assault survivors. *The Olympian*.
<https://www.theolympian.com/opinion/op-ed/article258066068.html>
- Scarlett, C. (2019, March 12). Creating a Spotify-powered app using Nuxt.js. *Smashing Magazine*.
<https://www.smashingmagazine.com/2019/03/spotify-app-vue-nuxt-javascript/>
- Scarlett, C. (2008). sIFRvault [Computer software].